WoSM Academy

MAT231-F27 · Calculus II

Double integrals

Rectangles, general regions, and Fubini's theorem

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Lecture 18 · Unit 18 · Fall 2027

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Last time

- Implicit function theorem: $F(\mathbf{x}, \mathbf{y}) = \mathbf{c}$ can be solved locally for $\mathbf{y}$ when $D_{\mathbf{y}}F$ is invertible.
- Inverse function theorem: $\det DF(\mathbf{a}) \neq 0$ gives a local inverse.
- This completes differential calculus in several variables.

Now integration. In one variable, $\int_a^b f(x)\,dx$ is the area under a graph. Today: the volume under a surface $z = f(x, y)$, and how to compute it with one-variable integrals.

This lecture

01 Double integrals over rectangles

02 Iterated integrals and Fubini's theorem

03 General regions

04 Applications

05 Summary and practice

Reading: Strang & Herman, *Calculus, Volume 3*, Sections 5.1 and 5.2.

01

Double integrals over rectangles

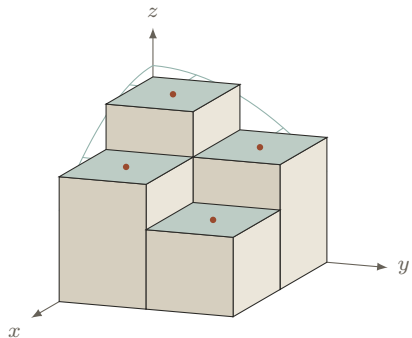
Volume under a surface

Let $f \geq 0$ on the rectangle $R = [a, b] \times [c, d]$.
To find the volume under its graph:

1. cut R into $m \times n$ small rectangles R_{ij}, each of area $\Delta A = \Delta x\, \Delta y$;
2. pick a sample point (x^*_{ij}, y^*_{ij}) in each;
3. add up the volumes of the boxes:

$$\sum_{i=1}^{m} \sum_{j=1}^{n} f(x^*_{ij}, y^*_{ij})\, \Delta A.$$

This is a **Riemann sum**.



$f = 8 - x^2 - y^2$ on $[0, 2]^2$, with 2×2 boxes at the midpoints.

The double integral

Definition

$\iint_R f(x,y)\,dA = \lim_{m,n\to\infty} \sum_{i=1}^{m}\sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*)\,\Delta A$, if the limit exists and is the same for every choice of sample points. Then f is **integrable** on R.

Theorem

A continuous function on R is integrable. So is a bounded function that is continuous except on finitely many smooth curves. (Proofs as in Calculus I, Units 18–19.)

For $f \geq 0$: the volume under the graph. In general: the volume above the xy-plane minus the volume below.

Example: Riemann sums

$f(x, y) = 8 - x^2 - y^2$ on $R = [0, 2] \times [0, 2]$.

- 2×2 squares ($\Delta A = 1$), upper right corners $(1, 1), (1, 2), (2, 1), (2, 2)$:

$$(6 + 3 + 3 + 0) \cdot 1 = 12.$$

- 2×2 squares, midpoints $(\frac{1}{2}, \frac{1}{2}), (\frac{1}{2}, \frac{3}{2}), (\frac{3}{2}, \frac{1}{2}), (\frac{3}{2}, \frac{3}{2})$:

$$(7.5 + 5.5 + 5.5 + 3.5) \cdot 1 = 22.$$

- 4×4 squares ($\Delta A = \frac{1}{4}$), midpoints: 21.5.

The exact value is $\frac{64}{3} \approx 21.33$ (shortly). The corners underestimate because f decreases in x and in y; the midpoints do much better, as in one variable.

Properties

For integrable f, g on R and constants α, β:

- **Linearity**: $\iint_R (\alpha f + \beta g)\,dA = \alpha \iint_R f\,dA + \beta \iint_R g\,dA$.
- **Monotonicity**: if $f \le g$ on R, then $\iint_R f\,dA \le \iint_R g\,dA$.
- **Area**: $\iint_R 1\,dA = \text{area}(R)$. So if $m \le f \le M$, then

$$m \cdot \text{area}(R) \le \iint_R f\,dA \le M \cdot \text{area}(R).$$

- **Additivity**: if R is cut into two rectangles R_1, R_2, then $\iint_R f = \iint_{R_1} f + \iint_{R_2} f$.

The **average value** of f on R is $\bar{f} = \dfrac{1}{\text{area}(R)} \iint_R f\,dA$.

02

Iterated integrals and Fubini's theorem

Computing by slicing

Fix x, and integrate over y:

$$A(x) = \int_c^d f(x,y)\,dy$$

is the area of the slice of the solid at x. Adding up the slices (as for volumes in Calculus I, Unit 24):

$$\text{volume} = \int_a^b A(x)\,dx = \int_a^b \Big(\int_c^d f(x,y)\,dy\Big)dx.$$

This is an **iterated integral**, usually written $\int_a^b \int_c^d f(x,y)\,dy\,dx$: the inner integral belongs to the inner differential. Slicing the other way gives $\int_c^d \int_a^b f(x,y)\,dx\,dy$.

Fubini's theorem

Theorem (Fubini)

If f is continuous on $R = [a,b] \times [c,d]$, then

$$\iint_R f\,dA = \int_a^b \int_c^d f(x,y)\,dy\,dx = \int_c^d \int_a^b f(x,y)\,dx\,dy.$$

Proof idea. Group a Riemann sum by columns:

$$\sum_i \Big(\sum_j f(x_i^*, y_j^*)\,\Delta y \Big) \Delta x.$$

The inner sum approximates $A(x_i^*) = \int_c^d f(x_i^*, y)\,dy$, and the outer one is then a Riemann sum for $\int_a^b A(x)\,dx$. Grouping by rows gives the other order.

Examples

Example (The volume from before)

$$\int_0^2 \int_0^2 (8 - x^2 - y^2)\, dy\, dx = \int_0^2 \Big(16 - 2x^2 - \frac{8}{3}\Big) dx = 32 - \frac{16}{3} - \frac{16}{3} = \frac{64}{3}.$$

Example (A product)

If $f(x, y) = g(x)\, h(y)$, the integral splits: $\iint_R g(x)h(y)\, dA = \Big(\int_a^b g\Big)\Big(\int_c^d h\Big)$. For instance

$$\int_0^1 \int_0^\pi x^2 \sin y\, dy\, dx = \Big(\int_0^1 x^2\, dx\Big)\Big(\int_0^\pi \sin y\, dy\Big) = \frac{1}{3} \cdot 2 = \frac{2}{3}.$$

Choose the order wisely

Example ($\iint_R x\,e^{xy}\,dA$, $R = [0,1] \times [0,1]$)

y **first**: for fixed x, an antiderivative of xe^{xy} in y is e^{xy}:

$$\int_0^1 \int_0^1 xe^{xy}\,dy\,dx = \int_0^1 (e^x - 1)\,dx = e - 2.$$

x **first**: $\int_0^1 xe^{xy}\,dx$ needs integration by parts, and then the y-integral is harder. Same answer, more work.

Fubini guarantees the two orders agree; it is our job to pick the easier one.

When Fubini fails

Example ($f(x,y) = \dfrac{x^2 - y^2}{(x^2+y^2)^2}$ **on** $[0,1] \times [0,1]$)

Since $\frac{\partial}{\partial y}\frac{y}{x^2+y^2} = f(x,y)$, $\displaystyle\int_0^1 \int_0^1 f\,dy\,dx = \int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4}$. By the symmetry $f(y,x) = -f(x,y)$, $\displaystyle\int_0^1 \int_0^1 f\,dx\,dy = -\frac{\pi}{4}$.

Check the hypothesis

f is not continuous on the closed square: it is unbounded near $(0,0)$, and not integrable. Then the two iterated integrals need not agree.

03

General regions

Integrating over a general region

Let D be a bounded region, inside a rectangle R. Extend f by zero:

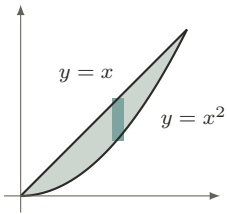
$$F(x,y) = \begin{cases} f(x,y), & (x,y) \in D, \\ 0, & (x,y) \in R \setminus D, \end{cases} \qquad \iint_D f\,dA = \iint_R F\,dA.$$

- If f is continuous on D and the boundary of D consists of finitely many smooth curves, F is integrable (its discontinuities lie on those curves).
- $\iint_D 1\,dA = \text{area}(D)$.
- Additivity: if $D = D_1 \cup D_2$ and D_1, D_2 overlap only along their boundaries, $\iint_D f = \iint_{D_1} f + \iint_{D_2} f$.

Type I and type II regions

Type I: $a \le x \le b,\ \ g_1(x) \le y \le g_2(x).$

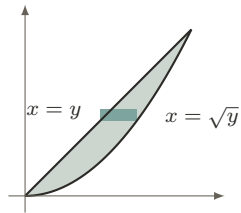
$$\iint_D f\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx$$



Vertical strips: y runs from the lower curve to the upper one.

Type II: $c \le y \le d,\ \ h_1(y) \le x \le h_2(y).$

$$\iint_D f\,dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\,dy$$



Horizontal strips: x runs from the left curve to the right one.

Example: both ways

D is the region between $y = x^2$ and $y = x$ (the picture), and $f(x, y) = xy$.

- **Type I**: $0 \le x \le 1$, $x^2 \le y \le x$:

$$\int_0^1 \int_{x^2}^{x} xy \, dy \, dx = \int_0^1 x \cdot \frac{x^2 - x^4}{2} \, dx = \frac{1}{2}\Big(\frac{1}{4} - \frac{1}{6}\Big) = \frac{1}{24}.$$

- **Type II**: $0 \le y \le 1$, $y \le x \le \sqrt{y}$:

$$\int_0^1 \int_{y}^{\sqrt{y}} xy \, dx \, dy = \int_0^1 y \cdot \frac{y - y^2}{2} \, dy = \frac{1}{2}\Big(\frac{1}{3} - \frac{1}{4}\Big) = \frac{1}{24}.$$

And $\text{area}(D) = \int_0^1 \int_{x^2}^{x} 1 \, dy \, dx = \int_0^1 (x - x^2) \, dx = \frac{1}{6}$.

Example: a volume

Example (The tetrahedron under $z = 1 - x - y$, above the triangle $x, y \geq 0$, $x + y \leq 1$)

The triangle is of type I with $0 \leq x \leq 1$, $0 \leq y \leq 1 - x$:

$$V = \int_0^1 \int_0^{1-x} (1 - x - y)\, dy\, dx = \int_0^1 \frac{(1-x)^2}{2}\, dx = \frac{1}{6}.$$

Check: a pyramid has volume $\frac{1}{3} \cdot \text{base} \cdot \text{height} = \frac{1}{3} \cdot \frac{1}{2} \cdot 1 = \frac{1}{6}$.

Setting up the limits

For a type I description

1. Sketch D. The outer limits are the smallest and largest x in D.
2. For fixed x, the vertical line enters and leaves D at the inner limits for y.
3. If the entry or exit curve changes formula, split D into pieces.

Example (The triangle with vertices $(0,0)$, $(2,0)$, $(1,1)$)

Type I needs two pieces (the top edge changes at $x = 1$); type II needs one:

$$\int_0^1 \int_0^x f \, dy \, dx + \int_1^2 \int_0^{2-x} f \, dy \, dx \;=\; \int_0^1 \int_y^{2-y} f \, dx \, dy.$$

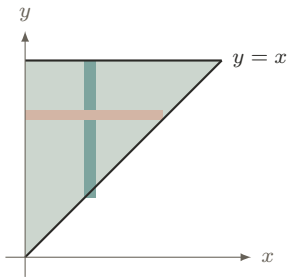
Changing the order of integration

$\int_0^1 \int_x^1 e^{y^2}\,dy\,dx$ — e^{y^2} has no elementary antiderivative.

The region: $0 \le x \le 1$, $x \le y \le 1$, the triangle $0 \le x \le y \le 1$. As a type II region: $0 \le y \le 1$, $0 \le x \le y$. So

$$\int_0^1 \int_0^y e^{y^2}\,dx\,dy = \int_0^1 y\,e^{y^2}\,dy = \frac{e-1}{2}.$$

The inner integral over x produced the factor y that the substitution $u = y^2$ needs.



Vertical strip: y from x to 1.
Horizontal strip: x from 0 to y.

04

Applications

Mass and centre of mass

A thin plate (**lamina**) occupying D, with density $\rho(x, y)$ (mass per unit area):

$$M = \iint_D \rho \, dA, \qquad \bar{x} = \frac{1}{M} \iint_D x \, \rho \, dA, \qquad \bar{y} = \frac{1}{M} \iint_D y \, \rho \, dA.$$

Example (The region under $y = x^2$, $0 \le x \le 1$, with $\rho = 1$)

$M = \int_0^1 x^2 \, dx = \frac{1}{3}, \quad \bar{x} = 3 \int_0^1 \int_0^{x^2} x \, dy \, dx = 3 \int_0^1 x^3 \, dx = \frac{3}{4},$
$\bar{y} = 3 \int_0^1 \int_0^{x^2} y \, dy \, dx = 3 \int_0^1 \frac{x^4}{2} \, dx = \frac{3}{10}.$

With $\rho = 1$ the centre of mass is the **centroid**: the average position of a point of D.

Probability

Random variables X, Y have a **joint density** $p(x,y) \geq 0$, with $\iint_{\mathbb{R}^2} p\,dA = 1$, if for regions D

$$P((X,Y) \in D) = \iint_D p(x,y)\,dA.$$

Example (X, Y independent and uniform on $[0,1]$)

Then $p = 1$ on the unit square and 0 outside, so probabilities are areas:

- $P(X+Y \leq 1) = \frac{1}{2}$ (a triangle);
- $P(Y \leq X^2) = \int_0^1 \int_0^{x^2} dy\,dx = \frac{1}{3}$.

Joint densities, expectations $E[g(X,Y)] = \iint g\,p\,dA$, and independence are developed in Probability Theory and Statistics.

05

Summary and practice

Summary

- $\iint_R f\,dA$: the limit of Riemann sums $\sum f(x_{ij}^*, y_{ij}^*)\,\Delta A$; for $f \geq 0$, the volume under the graph. Continuous functions are integrable.
- **Fubini**: for continuous f, the double integral is an iterated integral, in either order. Without continuity the orders can disagree.
- General regions: extend by zero. Type I: $\int_a^b \int_{g_1(x)}^{g_2(x)} \dots\, dy\, dx$; type II: $\int_c^d \int_{h_1(y)}^{h_2(y)} \dots\, dx\, dy$.
- Changing the order: sketch the region and describe it the other way. It can turn an impossible inner integral into an easy one.
- Area, average value, mass, centre of mass, probabilities.

Practice

1. Estimate $\iint_R xy\,dA$, $R=[0,2]^2$, by a midpoint sum with 4 squares, and compare with the exact value.
2. Compute $\int_0^1 \int_0^2 (x+y)^2\,dy\,dx$.
3. Compute $\iint_T (x+y)\,dA$, where T is the triangle with vertices $(0,0)$, $(1,0)$, $(0,1)$.
4. Find the area between $y=x^2$ and $y=2x$.
5. Compute $\displaystyle\int_0^{\pi}\int_x^{\pi} \frac{\sin y}{y}\,dy\,dx$ by changing the order.
6. Compute $\displaystyle\int_0^1\int_{\sqrt{x}}^1 \sin(y^3)\,dy\,dx$.
7. X, Y are independent and uniform on $[0,1]$. Find $P(XY \le \frac{1}{2})$.

Answers

1. $\frac{1}{4}+\frac{3}{4}+\frac{3}{4}+\frac{9}{4}=4$; exact: $\int_0^2 x\,dx\cdot\int_0^2 y\,dy=4$. (The midpoint rule is exact for xy.)
2. $\int_0^1\left(2x^2+4x+\frac{8}{3}\right)dx=\frac{16}{3}$.
3. $\int_0^1\int_0^{1-x}(x+y)\,dy\,dx=\frac{1}{6}+\frac{1}{6}=\frac{1}{3}$.
4. The curves meet at $x=0$ and $x=2$: $\int_0^2(2x-x^2)\,dx=\frac{4}{3}$.
5. $\int_0^{\pi}\int_0^{y}\frac{\sin y}{y}\,dx\,dy=\int_0^{\pi}\sin y\,dy=2$.
6. The region is $0\le y\le 1$, $0\le x\le y^2$: $\int_0^1 y^2\sin(y^3)\,dy=\frac{1-\cos 1}{3}$.
7. $1-\iint_{xy>1/2}1\,dA=1-\int_{1/2}^1\left(1-\frac{1}{2x}\right)dx=\frac{1+\ln 2}{2}\approx 0.847$.

Next time

Lecture 19 — Triple integrals (Unit 19)

- Integrals over boxes, and Fubini's theorem in three variables
- Integrals over general solids: choosing the order and the limits
- Mass, centre of mass, and probability in three dimensions

Reading: Strang & Herman, *Calculus, Volume 3*, Sections 5.4 and 5.6.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.